

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 1D nr. 21.10

$$A(z) = z^3 + 3z^2 + 4z + 2$$

$$A(-1) = -1 + 3 - 4 + 2 = 0 \Rightarrow z_1 = -1$$

$$\begin{array}{c|cccc} & 1 & 3 & 4 & 2 \\ -1 & \downarrow & -1 & -2 & -2 \\ \hline & 1 & 2 & 2 & 0 \end{array}$$

$$A(z) = (z + 1)(z^2 + 2z + 2) \rightarrow D = 2^2 - 4 \cdot 2 = -4$$

$$z_{2,3} = \frac{-2 \pm 2i}{2} \Rightarrow z_2 = -1 + i, z_3 = -1 - i$$

$$z_{1,2,3} \in \{-1, -1 + i, -1 - i\}$$

References